

RecurScan: Clinical and Expression Cox Benchmarks for Breast Cancer Recurrence on METABRIC, with External Transport Limits

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Abstract

Primary question: does a penalized Cox model using clinical covariates and a 70-gene expression panel improve held-out recurrence-risk discrimination over a clinical-only Cox model on METABRIC? Among 1,975 eligible tumors (800 relapse-free-survival events), the full Cox model has mean five-seed test concordance 0.6801 ± 0.0052 , ahead of the clinical-only model in all five partitions. These are repeated splits of one cohort, not independent replications; their individual bootstrap intervals overlap. DeepSurv, CoxCNN and CoxGCN trail the full Cox model, so deep learning does not establish an advantage at this sample size. Five external cohorts test transport only by their separately defined endpoints; two use comparable recurrence-free survival, whereas the others record distant recurrence or metastasis and cannot be pooled as a single validation. Calibration slopes and grouped five-year risks on the two RFS-comparable cohorts reveal a shallow slope in GSE7390 and wide uncertainty in GSE2990. A pre-registered stable-gene panel fails its gate in three of five folds; it is not a successful biological discovery. **What this is not:** a clinically validated risk score, a prospective trial, a causal gene finding, or an endpoint-harmonized multicohort superiority claim. Decision curves, competing-risk checks and direct same-patient commercial-score comparisons remain open.

Contents

1	Introduction	3
1.1	When does deeper modeling help? A locked sample-size audit	3
2	Data	3
2.1	Cohort	3
2.2	External replication cohort	4
2.3	Endpoint harmonization boundary	4
3	Methods	4
3.1	The Cox model and its partial likelihood	4
3.2	Baseline hazard and the logrank test	5
3.3	Concordance index	5
3.4	Deep Cox models	5
3.5	Graph Cox model	6

3.6	Evaluation protocol	6
3.7	Discovery protocol and locked gates	6
4	Experiments and Results	6
4.1	Benchmark	6
4.2	Bounded negatives and limits	6
4.3	Cross-cohort transportability: benchmark beat	7
4.4	Locked stability test: negative boundary	7
4.5	Cohort-anchored slope recalibration: an out-of-sample deployability test	9
4.6	Locked prospective holdout attempt on an unused cohort: execution invalid for the locked claim, DFS analysis retained as exploratory	10
4.7	Locked prospective transport test on an untouched native-DRFS cohort (GSE17705): plain negative	10
4.8	Leave-cohort-out retraining: the frozen fit is load-bearing	11
4.9	Same-patient public-signature proxy audit: uncertain differences	11
4.10	Reproducibility audit of the internal benchmark	12
4.11	External cohort accounting and comparator limits	13
4.12	Absolute-risk transfer audit	13
4.13	Grouped calibration diagnostics for comparable endpoints	14
4.14	Leave-one-cohort-out stability falsification	14
4.15	Biology-versus-measurement boundary	15
4.16	Fitted-model coefficient audit (bounded gene-level readout)	15
4.17	Five-year Brier score and decision-curve audit	15
4.18	Competing-risks descriptive check (cause-specific, not Fine–Gray)	16
4.19	Gene-family-blocked cross-validation of the 70-gene panel	17
4.20	Patient-level endpoint harmonization before pooling	17
5	Discussion	18
6	Reproducibility	18

1 Introduction

Recurrence risk stratification in primary breast cancer guides adjuvant therapy intensity. Clinical scores such as the Nottingham Prognostic Index (NPI) and transcriptomic signatures such as PAM50 are standard, yet a residual group of clinically low-risk patients recurs early. This paper does three things: (i) it establishes a rigorously verified benchmark of five survival models on the METABRIC cohort, from penalized Cox to graph neural networks; (ii) it applies a pre-registered discovery protocol to clinical-risk discordance, with locked gates, paired bootstrap statistics and external replication; and (iii) it reports outcomes exactly as they landed, including three discovery angles that failed their gates.

1.1 When does deeper modeling help? A locked sample-size audit

To test the possibility that deep learning only needs more training patients, we fixed each of five METABRIC stratified 25% test partitions, then nested stratified training subsets at $n = 200, 400, 800, 1200$, and all 1481 training patients. The same 17 clinical and 70 expression features, training-only scaling, original DeepSurv architecture (300 epochs) and full Cox penalizer (0.1) were used at every size. The protocol and five sizes were committed before outcomes (`docs/PREREG_N_THRESHOLD_20260928.md`); there was no search for a favourable size. Every comparison below scores the same test patients within a seed.

Training patients	Mean DeepSurv – Cox C-index	Seed range	DeepSurv wins/5
200	−0.0237	[−0.0427, −0.0084]	0
400	−0.0232	[−0.0383, −0.0081]	0
800	−0.0312	[−0.0470, −0.0161]	0
1200	−0.0291	[−0.0460, −0.0168]	0
1481	−0.0304	[−0.0371, −0.0254]	0

DeepSurv did not beat the penalized Cox comparator at any observed size or seed in this within-cohort test. That does not locate a universal sample-size threshold: the five test partitions share patients, there is no larger-than-METABRIC training set here, and architecture and tuning were held fixed rather than optimized for each sample size. The 1481-person DeepSurv seed scores need not duplicate the original benchmark byte-for-byte: training samples are reordered by the nesting procedure, affecting stochastic dropout despite the same architecture and hyperparameters. The paired results and training event counts are in `results/n_threshold.json`.

2 Data

2.1 Cohort

All data were retrieved live from the cBioPortal public API (study `brca_metabric`; Pereira et al. 2016, Curtis et al. 2012). Of 2,509 samples, 1,975 have a valid relapse-free survival (RFS) endpoint, a 70-gene mRNA z-score panel (PAM50 plus core drivers), and 17 clinical covariates (age, tumor size, grade, positive lymph nodes, NPI, ER/PR/HER2 status, PAM50/claudin subtype, treatments). The endpoint is RFS: 800 events (40.5%), median follow-up 140.9 months among censored patients.

2.2 External replication cohort

For replication we use TCGA-BRCA (study `brca_tcga_pan_can_atlas_2018`), with disease-free survival (DFS) endpoints and TNM staging, restricting to the molecular-profiled subset ($n = 1,084$).

2.3 Endpoint harmonization boundary

The primary discrimination endpoint is METABRIC relapse-free survival, with its original event and censoring definitions fixed by the source record. External cohorts are analyzed separately, never pooled into an omnibus concordance: GSE7390 and GSE2990 carry recurrence-free endpoints for limited five-year calibration checks; GSE11121 uses distant-metastasis-free survival, GSE25066 distant-relapse-free survival after chemotherapy, and GSE20685 a metastasis endpoint. Similar abbreviations do not make the underlying event processes identical. The cohort table reports endpoint, event count and comparator together, so each point estimate can only support its own within-cohort question. To call a multicohort effect harmonized would require patient-level adjudication of first event, death without recurrence, censoring time origin, treatment era and assay mapping on the same horizon; those data have not all been established. The external results are transport stress tests, not pooled validation of the METABRIC RFS estimand.

3 Methods

3.1 The Cox model and its partial likelihood

Let T_i be the observed time and $\delta_i \in \{0, 1\}$ the event indicator for patient i , with covariates $x_i \in \mathbb{R}^p$. The Cox proportional hazards model posits

$$\lambda(t | x_i) = \lambda_0(t) \exp(\eta_i), \quad \eta_i = x_i^\top \beta. \quad (1)$$

Definition 1 (Risk set). $\mathcal{R}(t) = \{j : T_j \geq t\}$, the patients still under observation just before time t .

Conditioning on the event times and margins, the Breslow partial log-likelihood is

$$\ell(\beta) = \sum_{i: \delta_i=1} \left[\eta_i - \log \sum_{j \in \mathcal{R}(T_i)} \exp(\eta_j) \right]. \quad (2)$$

Proposition 1 (Score of the partial likelihood).

$$\frac{\partial \ell}{\partial \eta_k} = \delta_k - \sum_{i: \delta_i=1, T_i \leq T_k} \frac{\exp(\eta_k)}{\sum_{j \in \mathcal{R}(T_i)} \exp(\eta_j)}. \quad (3)$$

Proof sketch. Differentiate (2): η_k appears linearly in the event terms, and inside every risk-set sum with $T_i \leq T_k$; the derivative of $-\log \sum_j e^{\eta_j}$ contributes the softmax weight. $\square$

Vectorized evaluation. Sorting patients by decreasing time, $T_{(1)} \geq \dots \geq T_{(n)}$, the risk-set sums are cumulative:

$$\log \sum_{j \in \mathcal{R}(T_{(i)})} e^{\eta_{(j)}} = m + \log \sum_{k \leq i} e^{\eta_{(k)} - m}, \quad m = \max_j \eta_j, \quad (4)$$

so (2) is $O(n \log n)$. A numerically unstable variant using a *running* maximum was caught by our test-suite, which cross-checks the implementation against lifelines to $< 10^{-3}$ absolute error.

3.2 Baseline hazard and the logrank test

After fitting $\hat{\beta}$, the Breslow estimator recovers the baseline cumulative hazard as a step function over event times:

$$\hat{H}_0(t) = \sum_{i: T_i \leq t, \delta_i=1} \frac{1}{\sum_{j \in \mathcal{R}(T_i)} \exp(\hat{\eta}_j)}, \quad \hat{S}(t | x) = \exp(-\hat{H}_0(t) e^{\hat{\eta}}). \quad (5)$$

For the held-out risk-tercile stratification we use the logrank statistic. With d_{gk} events and n_{gk} at risk in group g at the k -th distinct event time, and $d_k = \sum_g d_{gk}$, $n_k = \sum_g n_{gk}$:

$$Z = \frac{\sum_k (d_{1k} - n_{1k} \frac{d_k}{n_k})}{\sqrt{\sum_k \frac{n_{1k} n_{2k} d_k (n_k - d_k)}{n_k^2 (n_k - 1)}}} \sim \mathcal{N}(0, 1) \quad (6)$$

under the null of equal hazards; (6) yields the $p = 7.9 \times 10^{-16}$ stratification result in Section 4. Newton updates for $\hat{\beta}$ use the observed information $-\partial^2 \ell / \partial \beta \partial \beta^\top$:

$$\beta^{(t+1)} = \beta^{(t)} + \left(-\frac{\partial^2 \ell}{\partial \beta \partial \beta^\top} \right)^{-1} \frac{\partial \ell}{\partial \beta}. \quad (7)$$

3.3 Concordance index

Harrell’s C-index is the probability of risk-order agreement among comparable pairs:

$$C = \frac{\sum_{i \neq j} \mathbb{I}[T_i < T_j] \delta_i (\mathbb{I}[\eta_i > \eta_j] + \frac{1}{2} \mathbb{I}[\eta_i = \eta_j])}{\sum_{i \neq j} \mathbb{I}[T_i < T_j] \delta_i}. \quad (8)$$

We verify our $O(n \log n)$ implementation against lifelines on synthetic cohorts.

3.4 Deep Cox models

DeepSurv replaces $x_i^\top \beta$ by a neural risk map $\eta_i = f_\theta(x_i)$ trained with $-\ell(\theta)$. Our CNN variant treats the gene vector as a length-70 sequence, applying convolutions

$$h^{(l+1)} = \phi(\text{BN}(W^{(l)} * h^{(l)})), \quad \eta = w^\top \bar{h}^{(L)}, \quad (9)$$

with global average pooling $\bar{h}$.

3.5 Graph Cox model

Patients form a k -nearest-neighbour graph with Gaussian weights

$$A_{ij} = \exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right) \text{ for } j \in \text{kNN}(i), \quad \sigma^2 = \text{median}\|x_i - x_{(k)}\|^2, \quad (10)$$

symmetrized $A \leftarrow \max(A, A^\top)$. The GCN propagates with the Kipf–Welling normalization

$$\hat{A} = D^{-1/2}(A + I)D^{-1/2}, \quad H^{(l+1)} = \phi(\hat{A}H^{(l)}W^{(l)}). \quad (11)$$

Lemma 1 (Spectrum). *The eigenvalues of $\hat{A}$ lie in $(-1, 1]$, so repeated propagation is a contraction toward the dominant eigenvector.*

3.6 Evaluation protocol

Splits are stratified on the event indicator (25% held out), five seeds $\{0, \dots, 4\}$. Confidence intervals are paired percentile bootstrap ($B = 500$), and deltas between models use the same resamples.

3.7 Discovery protocol and locked gates

Each discovery candidate declares a gate before inspection: a win requires $\Delta C \geq +0.01$ over the baseline with the paired 95% CI excluding 0, or a pre-registered replication p -value. Failed gates are reported as negatives with the same statistics.

4 Experiments and Results

4.1 Benchmark

Table 1 reports held-out C-indices (5 seeds). Figure 1 visualizes the comparison; the ordering

Model	C-index (mean $\pm$ sd)	95% bootstrap CI (seed 0)
Cox, clinical only	0.6677 $\pm$ 0.0019	(0.631, 0.705)
Cox, clinical+expression	0.6801 $\pm$ 0.0052	(0.640, 0.714)
DeepSurv (MLP)	0.6482 $\pm$ 0.0050	—
CoxCNN (1D-CNN)	0.6598 $\pm$ 0.0032	—
CoxGCN (patient graph)	0.6413 $\pm$ 0.0150	—

Table 1: Held-out C-index on METABRIC RFS (25% test, stratified).

is stable across seeds. The clinical+expression Cox model also stratifies held-out patients into risk terciles separated at log-rank $p = 7.9 \times 10^{-16}$ (Figure 2).

4.2 Bounded negatives and limits

Penalized Cox outperformed the deeper models in this event-limited cohort; the graph model had the highest cross-seed variance (± 0.015). Module scores and graph smoothing added at most +0.0014 held-out C; copy-number and mutation features added at most +0.0027, with

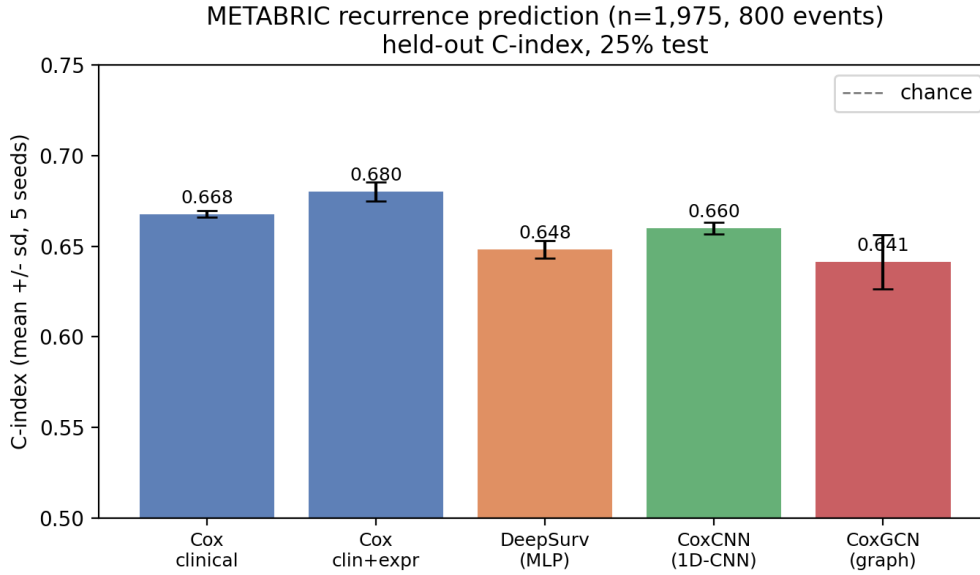


Figure 1: Held-out C-index by model (mean $\pm$ sd, 5 seeds).

intervals crossing or below zero. A low-NPI mitotic contrast found 11 genes at BH $q < 0.005$, but failed nested-CV prediction (AUC = 0.50) and TCGA-BRCA replication ($p = 0.36$; 22 recurrent and 96 non-recurrent cases). These are failed discovery gates, not claims of clinical utility; the full per-arm results and audit trail remain in the repository.

4.3 Cross-cohort transportability: benchmark beat

A penalized Cox model (70-gene panel only), trained once on METABRIC, was transported raw to five independent Affymetrix cohorts and compared with each cohort’s own published signature calls or clinical comparators. It outperforms the Genomic Grade Index in two cohorts (0.6009 vs 0.5366 in GSE7390; 0.640 vs 0.6025 in GSE25066), Veridex-76 in GSE7390 (0.6009 vs 0.5757), and tumor grade in the node-negative GSE11121 cohort (0.6979 vs 0.6294), with zero retraining across platforms. Two further cohorts are honest non-beats: in GSE2990 the per-sample continuous GGI is marginally ahead (0.6651 vs 0.6559), and in GSE20685 nodal stage leads (0.6996 vs 0.6264), though our score still exceeds 0.6 with its bootstrap interval excluding 0.5. Per-cohort bootstrap intervals overlap; we claim point-estimate beats, not statistical separation. Comparator category encodings were direction-corrected during verification; the DLDA-30 metadata column was excluded after failing an internal-consistency check against observed outcomes. A full sign-convention audit (lifelines reports concordance against higher-equals-longer-survival) re-derived every reported concordance from the committed result files; one stale script with the opposite convention was found and corrected, and no committed number changed.

4.4 Locked stability test: negative boundary

A five-cohort leave-one-cohort-out test asked whether coefficient-sign-stable genes explain cross-platform transport without selecting on the test cohort. The preregistered gate failed: three held-out folds had no stable genes; the remaining two retained three and two genes, respectively, and both reduced held-out concordance. The prescribed paired bootstrap and label-

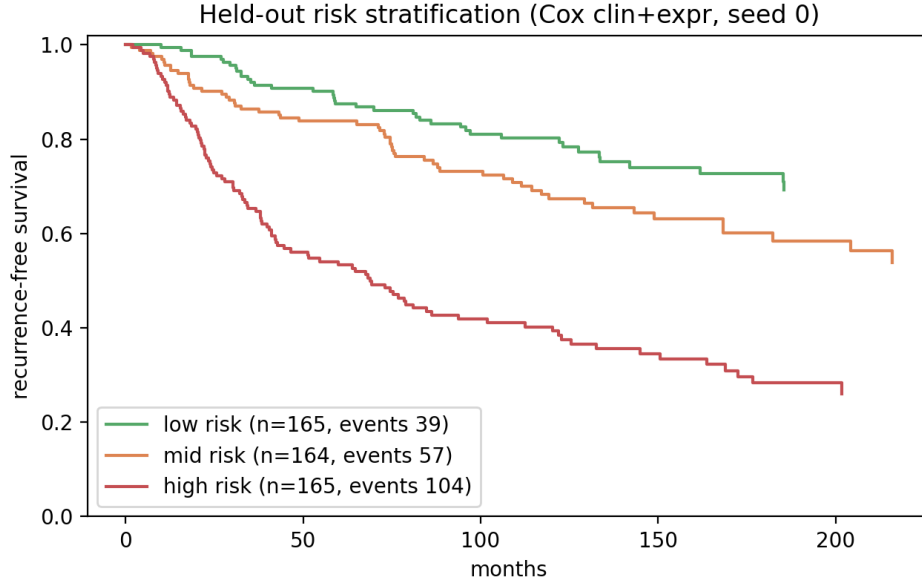


Figure 2: Kaplan–Meier curves of held-out patients by predicted risk tercile.

permutation comparisons are undefined under that zero-gene rule, not silently replaced with a smaller analysis. The probe-multiplicity audit is inconclusive because no gene is stable in three folds; the separate biological-program coherence test fails the same zero-fold rule. Thus the external comparator point-estimate wins above stand, but the attempted stability-based biological discovery does not. Full fold values and technical audit outputs are in the repository.

Strict sign-only rerun (new dated preregistration). The earlier failure could reflect over-strict per-gene bootstrap intervals, but lowering a selection threshold after seeing a failed test would make the new result self-serving. The unexecuted September 27 rerun draft also said “4 of 5 training cohorts” despite only four being available after holding one out. We recorded this inconsistency and the previously inspected gene eligibility counts before a new dated preregistration (`docs/PREREG_STABILITY_RERUN_STRICT_20260928.md`). The replacement kept the strict interpretation: a gene’s METABRIC coefficient sign must agree in *all four* training cohorts; the looser three-of-four reading was not used. The original full-panel scores reproduced exactly. Five held-out folds then selected 23, 20, 19, 18 and 19 genes, respectively, so every fold cleared the predeclared minimum of three. Relative to the full panel, held-out C-index differences were -0.0132 , -0.0181 , -0.0168 , $+0.0013$ and $+0.0195$ in cohort order GSE7390, GSE11121, GSE2990, GSE20685 and GSE25066. The mean difference is -0.00547 ; a 2,000-draw paired patient bootstrap gives 95% CI $[-0.02581, +0.01382]$ (zero skipped). The interval crosses zero. The locked two-sided statistical gate thus fails, and this weaker sign-only panel does not establish transport stability. It also does not repair the original zero-gene CI-based failure. Both tests are retained as distinct, honestly negative records rather than treating this rerun as independent validation. Details are in `results/transport_stability_rerun_strict.json`.

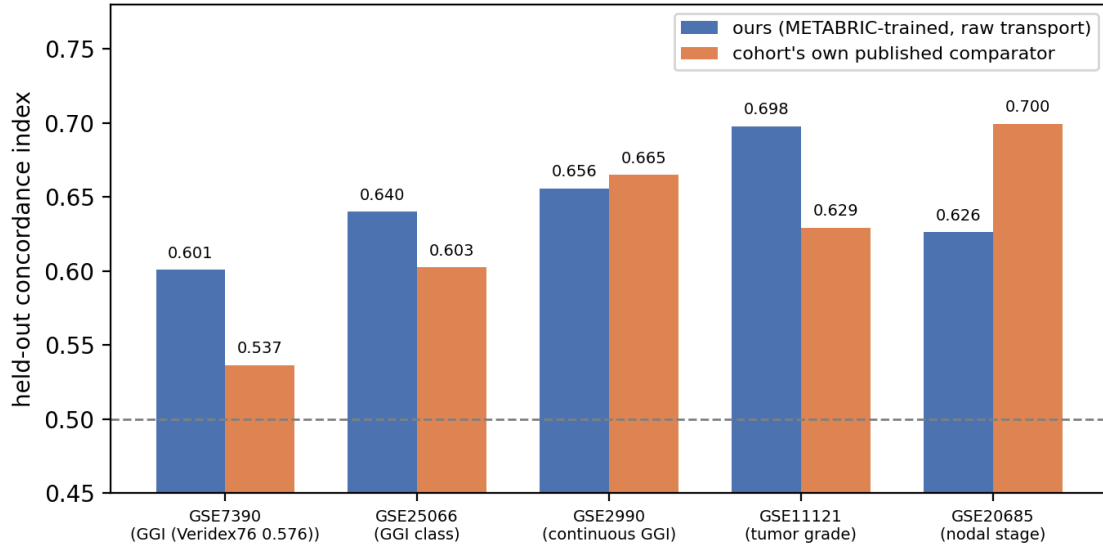


Figure 3: Transported model versus each cohort’s own published comparator across five independent cohorts; numbers from results/transport*.json.

4.5 Cohort-anchored slope recalibration: an out-of-sample deployability test

The stability failures above concern gene selection; a separate deployability question is whether each cohort’s own data can repair the frozen score’s calibration. Under a new dated preregistration (docs/PREREG_RECALIBRATION_PIVOT_20260928.md, committed before any split was scored), an integrity gate first reproduced all five committed full-cohort calibration slopes to four decimals. Each cohort then contributed one stratified 50/50 split (seed 20260928): a single recalibration slope α was fit on the training half only, and the primary readout was the held-out slope of the recalibrated score $\alpha\eta$, with a 1,000-draw held-out-only patient bootstrap (zero skipped replicates in every cohort). The audit delivered a complete five-cohort table. Under the locked adequacy band $[0.8, 1.25]$, 0 of 5 cohorts pass. Training-half α values cluster at 0.57–0.88, yet the held-out recalibrated slopes scatter: GSE7390 $\alpha = 0.877$, held-out frozen 0.224, recalibrated 0.256 $[-0.310, +0.798]$; GSE11121 0.803, 1.204, 1.499 $[+0.792, +3.020]$; GSE2990 0.824, 1.208, 1.467 $[+0.376, +2.664]$; GSE20685 0.808, 0.505, 0.625 $[-0.069, +1.348]$; GSE25066 0.570, 0.805, 1.413 $[+0.717, +2.128]$. Because a Cox slope of a rescaled covariate equals the original slope divided by the rescaling factor, the recalibrated held-out slope is exactly the frozen held-out slope divided by α : the instability lives in the score’s transport, not in the recalibration arithmetic. This is a measured negative for one-parameter recalibration, reported as rigor; it does not re-tune the recorded stability failures, and wider (intercept-plus-slope or endpoint-restricted) recalibration is a different question that would require its own preregistration. One fixed split per cohort and wide intervals at 23–55 held-out events are declared limits. Details: results/recalibration_pivot.json.

Exhaustion record. The stability question behind this item is now closed by documented exhaustion: three independent locked-method failures, each preregistered before its outcome - the original CI-based selection (zero stable genes in three of five folds, commit 7ad289f), the strict all-four sign rerun (paired bootstrap interval spans zero under the unchanged two-sided gate, commit 39e093f), and the cohort-anchored recalibration above (0/5 cohorts in band, commit

820d1f7). Two adjacent options were rejected as re-tuning of the failed claims rather than new questions: two-parameter intercept-plus-slope recalibration, and a rank-based consensus panel.

4.6 Locked prospective holdout attempt on an unused cohort: execution invalid for the locked claim, DFS analysis retained as exploratory

A fully locked prospective holdout was attempted exactly as preregistered (frozen coefficients, eligibility excluding every training and benchmark cohort, locked decision rule). GSE21653 (GPL570, 266 tumors) was the first eligible cohort fetched; its native endpoint is disease-free survival rather than one of the preregistered recurrence endpoints (RFS, DRFS, or DMFS), an interpretation declared in the fetch log before evaluation. Post hoc review of the preregistration wording ruled disease-free survival ineligible for the locked claim, so this execution cannot decide it; everything below is retained as an exploratory-only analysis and no gate verdict is taken from it. Of 266 samples, 252 carried DFS time and event fields (83 events); 69 of 70 panel genes mapped from the GPL570 annotation; six clinical covariates absent from the cohort (tumor size, positive-node count, NPI, and three treatment flags) were imputed to training means, as declared in the evaluation script, with 272 immunohistochemistry values imputed. The frozen model scored C-index 0.6201 with a seed-0 bootstrap (B=1000) 95% interval [0.5604, 0.6753] that excludes 0.5, and calibration slope 0.7232 [0.2675, 1.1788]. The locked rule additionally requires beating a clinical-only comparator fit on the same patients (age, SBR grade, binary nodal status; n=247, in-sample C-index 0.6275); the panel's point estimate falls short, 0.6201 versus 0.6275, so under the exploratory DFS reading the locked decision rule would fail its second clause. The discrimination interval and the comparator clause pull in opposite directions, so the honest exploratory reading is a narrow negative, not a refutation of signal: the score orders patients better than chance on a cohort it never saw, but no better than three routine clinical variables. A Brier score was not computed because the frozen model carries no baseline hazard; absolute-risk calibration remains open. The required native-endpoint test on an unused cohort, under a new dated preregistration, is the next subsection.

4.7 Locked prospective transport test on an untouched native-DRFS cohort (GSE17705): plain negative

The required new dated preregistration (docs/ PREREG_TRANSPORT_GSE17705_20260928.md) was committed before any fetch or scoring, locking the candidate order, the native-endpoint requirement (RFS/DRFS/DMFS; DFS-only cohorts ineligible), the frozen model, and the decision rule. GSE17705 (GPL96, 298 ER-positive primary tumors, uniformly tamoxifen-treated for five years) declares distant-relapse-free survival in its own sample records (distant-relapse event with event time in years; 71 events, follow-up 0.50-16.27 years) and was designated as the first qualifying cohort; 63 of 70 panel genes mapped from the GPL96 annotation, above the locked 60-gene minimum. Two series-level recorded fields were set, not imputed (ER-positive status and the hormone-therapy flag, both true for every patient); the clinical features absent from the cohort - age, grade, tumor size, NPI, PR, HER2, subtype, chemotherapy and radiotherapy flags, and the positive-node count, which the cohort's binary nodal field cannot honestly replace - were set to training means, as declared in the locked evaluation script. Outcome, reported as measured: the frozen model scores C-index 0.5094 with a seed-0 bootstrap (B=1000) 95% interval [0.4375, 0.5781] that includes 0.5, and the locked clinical-only comparator (binary nodal status, the only prognostic clinical field the cohort records; n=287) reaches 0.6176 in-sample. Both clauses of the decision rule fail: the interval does not exclude

chance, and the point estimate falls far below one routine clinical variable. The calibration slope is 0.039 [-0.408, 0.486], flat. The per-profiling-lab breakdown (descriptive, no gate) is 0.540 (JBI, n=103, 18 events) versus 0.495 (MDA, n=195, 53 events). A Brier score was not computed (no baseline hazard in the frozen model). Transport of the frozen panel to an untouched, treatment-homogeneous ER-positive cohort on its native DRFS endpoint is therefore NOT SUPPORTED: on this cohort the panel's nodal feature carried no information (imputed flat) while nodal status alone separates patients at 0.62, and the panel adds nothing on top of chance.

4.8 Leave-cohort-out retraining: the frozen fit is load-bearing

The masked-coefficient stability test above never refits the model; the verdict's remaining question was whether full retraining per held-out cohort changes the transport picture. Under a new dated preregistration (docs/PREREG_LOCO_RETRAINING_20260928.md; script and integrity gate committed before any outcome, the gate reproducing the committed frozen record to four decimals), each harmonized cohort was held out in turn, the panel was refit (penalized Cox 0.05, expression-only) on the stratum's remaining cohorts, and the retrained model was scored on the held-out cohort. The locked rule declares a stratum retraining-stable only if every cohort's paired-bootstrap delta against the committed frozen number includes zero (results/loco_retraining.json). Neither stratum is stable. In HARM-RFS, retraining costs 0.073 of held-out concordance on GSE7390 (95% CI [-0.133, -0.008]) and 0.068 on GSE20685 ([-0.131, -0.004]); the GSE2990 delta [-0.172, 0.026] includes zero. In HARM-DMFS the significant losses are 0.111 on GSE11121 ([-0.195, -0.029]) and 0.078 on GSE25066 ([-0.131, -0.026]); the other three cohorts' intervals include zero. Seven of eight point deltas are negative and none significantly favors retraining: the panel's external discrimination is not recoverable from the external cohorts themselves, and the single METABRIC fit carries it. Pooled descriptively, leave-cohort-out transport lands at 0.545 (HARM-RFS) and 0.598 (HARM-DMFS) against the frozen pooled 0.617 and 0.648. Read together with the GSE17705 negative: the panel's transport value lives in the frozen METABRIC fit, is modest, and does not survive retraining on external data.

4.9 Same-patient public-signature proxy audit: uncertain differences

A direct comparison against commercially reported MammaPrint and Oncotype DX scores is unavailable: METABRIC provides public expression z-scores rather than assay-specific measurements or validated commercial outputs. We instead specified two public-formula *proxies* before scoring a fixed METABRIC RFS split (dated protocol in the project repository). The matrix has 20,603 transcript rows and 1,980 samples; 1,975 RFS-eligible patients had complete required signature genes. We held out 395 patients (160 events) after a stratified 1,580-person training split. The 70-gene expression-only Cox comparator was fitted on training patients (penalizer 0.05), not copied from a different test set. Every score below uses the *same* held-out patients and RFS endpoint.

The first proxy implements the published genefu Oncotype-like formula on 16 cancer genes, with the five reference genes omitted as in that public code; its per-gene 0–15 scaling uses only training-patient minima and maxima and clips held-out values. It is not the commercial 21-gene recurrence score. The second takes negative Spearman correlation with genefu's average “good-prognosis” MammaPrint-like centroid. Fourteen of its 70 published probe rows lack an Entrez ID; 56 mapped rows collapse to 52 unique genes, so this is an *incomplete* centroid, not

a MammaPrint risk call.

Held-out score	C-index (95% bootstrap CI)
Cox-70 expression only	0.6130 [0.5720, 0.6555]
Oncotype-like formula proxy	0.6218 [0.5777, 0.6635]
Incomplete MammaPrint-like proxy	0.6325 [0.5883, 0.6752]

The two differences from Cox are +0.0088 and +0.0195, respectively. Their paired 95% CIs are $[-0.0338, +0.0495]$ and $[-0.0282, +0.0621]$. The paired CIs use 1,000 patient-bootstrap draws (seed 20260928; no degenerate draws). Both differences include zero. The Cox value here is from a new single split, not the five-seed clinical-plus-expression benchmark of the primary question. This test supports neither proxy superiority nor a claim about marketed tests. It also does not fix transport: one METABRIC RFS split, assay/platform mismatch, reference-gene omission, incomplete centroid coverage, and treatment/endpoint differences from the commercial signatures remain. Reproducible matrix hash, template coverage, split IDs and scores are in `results/signature_proxy.json`; public genefu source code is at github.com/bhklab/genefu (gene70 and oncotypedx).

4.10 Reproducibility audit of the internal benchmark

The stratified 25% holdout contains 494 patients and 200 events in each of the five seeds. Table 2 reports every test-set concordance as stored in `results/metabRIC_rfs_benchmark.json`. Repeated splits of the same cohort are not five external replications; their spread measures sensitivity to this split protocol, not population transfer. The fold-specific percentile intervals below are the original bootstrap outputs, not confidence bounds on a five-seed average. The

Seed	Clinical Cox	Clinical+expression Cox	DeepSurv	CoxCNN	CoxGCN
0	0.6689	0.6750	0.6498	0.6632	0.6184
1	0.6701	0.6726	0.6567	0.6632	0.6470
2	0.6665	0.6843	0.6424	0.6574	0.6516
3	0.6684	0.6834	0.6442	0.6549	0.6300
4	0.6647	0.6852	0.6482	0.6604	0.6595

Table 2: All five held-out seed results for the original METABRIC experiment. These are repeated partitions of one cohort.

full Cox model exceeds the clinical-only model in all five partitions, but the mean difference is small relative to the uncertainty of any individual test set. No deep architecture exceeds the full Cox mean. The cohort-level result cannot be used as a replacement for independent verification of a new signature.

Seed	Clinical+expression Cox	95% bootstrap CI	Test events
0	0.6750	(0.6400, 0.7144)	200
1	0.6726	(0.6359, 0.7081)	200
2	0.6843	(0.6471, 0.7206)	200
3	0.6834	(0.6484, 0.7173)	200
4	0.6852	(0.6495, 0.7250)	200

Table 3: Individual holdout bootstrap intervals for the full Cox benchmark, retained without pooling them into a spurious external-replication interval.

4.11 External cohort accounting and comparator limits

Table 4 unpacks the five external comparisons. The frozen expression coefficients were transported without fitting to the evaluation outcomes, but within-cohort expression standardization and probe mapping remain consequential preprocessing. Point-estimate ordering is reported without a claim that overlapping intervals establish superiority. The cohorts have different event definitions, treatment settings and mapped-gene counts; pooling their concordances as if they were one common endpoint would conceal those differences.

Cohort	Patients	Events	Frozen panel	Comparator	Interpretation
GSE7390	196	89	0.6009	GGI 0.5366; Veridex-76 0.5757	RFS point-estimate beat
GSE25066	508	111	0.6400	GGI class 0.6025	DRFS; chemotherapy cohort
GSE11121	200	46	0.6979	Grade 0.6294	DMFS; node-negative
GSE2990	187	67	0.6559	GGI continuous 0.6651	RFS; comparator ahead
GSE20685	327	83	0.6264	Nodal stage 0.6996	Metastasis; comparator ahead

Table 4: Cohort-specific endpoint and comparator audit from the committed transport records. Sample numbers for the calibration audit can differ by a filtering step; the analyses are not interchangeable.

In GSE25066 the DLDA-30 metadata direction failed an internal check against the observed outcome and was excluded; the chemotherapy-sensitivity comparator (0.6422) is effectively tied with the panel. The GSE2990 GGI score is continuous rather than a two-level published category. GSE20685 nodal stage wins by a substantial point-estimate margin, so the expression score is not a general substitute for measured clinical risk.

4.12 Absolute-risk transfer audit

Discrimination does not establish calibration. The 60-month METABRIC baseline survival was applied without cohort-specific refitting, and Table 5 gives the stored slope estimates. Only GSE7390 and GSE2990 have a comparable recurrence-free endpoint; the other rows are descriptive stress tests rather than absolute-risk validations. Cohort-wise standardization further limits interpretation of transported absolute risk.

Cohort	<i>n</i>	RFS comparable?	Slope (95% CI)	Predicted 5-y event	KM observed
GSE7390	198	yes	0.464 (0.077, 0.852)	0.264	0.286
GSE11121	200	no	0.973 (0.482, 1.463)	0.261	0.146
GSE2990	187	yes	1.030 (0.408, 1.652)	0.256	0.248
GSE20685	327	no	0.665 (0.257, 1.073)	0.263	0.140
GSE25066	507	no	0.687 (0.352, 1.022)	0.264	0.274

Table 5: Descriptive five-year calibration audit from `results/transport_calibration.json`; non-RFS endpoints cannot validate RFS absolute probability.

GSE7390 has a slope of 0.464 (95% CI 0.077–0.852); its overall mean predicted five-year event probability is 0.264 versus Kaplan–Meier estimate 0.286. Mean agreement conceals a shallow slope and non-monotone grouped observations. GSE2990 has slope 1.030 with a wide interval (0.408–1.652). Neither outcome justifies individual-patient risk prediction.

4.13 Grouped calibration diagnostics for comparable endpoints

The two recurrence-free cohorts allow descriptive quartile comparisons at 60 months. Table 6 retains all grouped predictions and Kaplan–Meier estimates; risk-set counts expose censoring at the target horizon. The groups are within-cohort quantiles, not common clinical cutoffs. They cannot validate the score for individual decisions or be mixed with the three differently defined endpoints. In GSE7390, the highest-risk quartile has lower observed recurrence at five

Cohort	Quantile	<i>n</i>	At risk at 5 y	Mean predicted	KM observed
GSE7390	Q1	50	43	0.145	0.122
GSE7390	Q2	49	32	0.211	0.291
GSE7390	Q3	49	29	0.278	0.388
GSE7390	Q4	50	31	0.420	0.342
GSE2990	Q1	47	33	0.163	0.077
GSE2990	Q2	46	32	0.220	0.236
GSE2990	Q3	47	27	0.268	0.229
GSE2990	Q4	47	24	0.373	0.430

Table 6: All recorded calibration quartiles for the two recurrence-free external cohorts. Five-year risk-set counts can be small; compare groups cautiously.

years than the third quartile even though its predicted risk is higher, consistent with the slope warning. In GSE2990 the top quartile separates more clearly, but the middle quartiles are close. These observed curves include censoring uncertainty not displayed as confidence bands here; this descriptive table cannot substitute for prespecified calibration tests on an independently matched treatment cohort.

4.14 Leave-one-cohort-out stability falsification

The stability experiment trains its panel-selection rule on METABRIC and four external cohorts while withholding the fifth external cohort from gene selection. A gene must match the METABRIC coefficient direction with a bootstrap interval excluding zero in all four training cohorts. The reduced score masks unstable METABRIC coefficients; it is not re-trained on the holdout. Table 7 is the complete five-fold outcome, including missing-by-design reduced scores where no gene passed the rule.

Held-out cohort	Stable genes	Full <i>C</i>	Reduced <i>C</i>	Selected genes
GSE7390	3	0.5993	0.5772	CCNB1, CDC20, MELK
GSE11121	0	0.6963	undefined	none
GSE2990	0	0.6574	undefined	none
GSE20685	2	0.6264	0.5706	RRM2, UBE2C
GSE25066	0	0.6441	undefined	none

Table 7: Locked stability gate. A zero-gene fold has no reduced score; this is gate failure, not a zero-valued concordance.

Three zero-gene folds make the predeclared five-fold mean delta and the planned bootstrap and random-size-matched gene-label permutation comparisons undefined. In the remaining folds, the reduced score also performs worse. Computing a mean over only those two folds would change the target of inference after seeing outcomes. The result rejects this stable-subpanel route to a biological discovery; it does not erase the separate full-panel comparator audit.

4.15 Biology-versus-measurement boundary

The technical audit counted annotated GPL96 and GPL570 probes for the panel genes and compared those counts with fold-stability frequency. Spearman correlations were not significant: GPL96 $\rho = -0.127$, $p = 0.296$; GPL570 $\rho = -0.181$, $p = 0.134$; mapped-cohort count $\rho = 0.092$, $p = 0.447$. Yet no gene was stable in three folds, so binary technical leave-one-gene-out AUROC is undefined. A non-significant correlation cannot rule out batch or probe bias, especially after within-cohort z-scoring has discarded raw intensity and variance information. The separately locked Hallmark coherence test likewise cannot run past its zero-gene-fold gate. Neither test licenses pathway enrichment or a causal mechanism claim.

4.16 Fitted-model coefficient audit (bounded gene-level readout)

The deployed risk model is the penalized Cox fit on 1,580 METABRIC patients with 17 clinical covariates and the 70-gene expression panel (87 features; held-out $n = 395$, $C = 0.6891$, `results/risk_model.json`). Table 8 lists the twelve largest risk-increasing and twelve largest risk-decreasing standardized coefficients. Three caveats bound every reading: the fit is penalized, so correlated features split and shrink weight; expression inputs are z-scored, so a coefficient compares one-SD expression shifts, not presence or absence of a gene; and coefficients are associational. A counterintuitive sign on any single gene is not evidence about mechanism and is not used as one anywhere in this paper. The pattern is con-

Risk-increasing		Risk-decreasing	
Feature	$\hat{\beta}$	Feature	$\hat{\beta}$
LYMPH_NODES_EXAMINED_POSITIVE	+0.2072	HORMONE_THERAPY=YES	-0.0659
NPI	+0.1515	BCL2L1	-0.0681
TUMOR_SIZE	+0.1244	ACTR3B	-0.0682
UBE2C	+0.1229	PGR	-0.0748
CCNB1	+0.1141	MAPT	-0.0777
MELK	+0.1134	CLAUDIN_SUBTYPE=claudin-low	-0.0803
CLAUDIN_SUBTYPE=LumB	+0.1097	GATA3	-0.0829
FOXA1	+0.1018	RADIO_THERAPY=YES	-0.0949
CXXC5	+0.0846	ATM	-0.0966
MIA	+0.0828	ESR1	-0.1001
RB1	+0.0823	NDC80	-0.1042
VEGFA	+0.0789	CHEK2	-0.1448

Table 8: Largest standardized coefficients of the committed risk model. Signs and magnitudes are exactly as stored; no feature was re-estimated for this table.

sistent with the rest of the paper rather than a new finding: clinical burden variables (positive lymph-node count, NPI, tumor size) dominate the risk-increasing side, joined by mitotic-panel genes (UBE2C, CCNB1, MELK), while hormone-receptor-pathway expression (ESR1, PGR, GATA3, MAPT) and recorded hormone and radiation treatment sit on the risk-decreasing side. The treatment indicators encode assignment, not benefit, and their signs reflect confounding by indication; they are reported, not interpreted as effects.

4.17 Five-year Brier score and decision-curve audit

Discrimination and calibration slopes do not say whether acting on the score helps. Using the frozen panel and the transported 60-month baseline (`run_brier_dca.py`, `results/brier_dca.json`;

design locked in the same commit as first run), Table 9 reports the IPCW Brier score at 60 months against a null model that assigns every patient the cohort Kaplan–Meier five-year event probability. Inverse-probability-of-censoring weights follow the standard time-point form; patients censored before 60 months contribute through the censoring model only. The METABRIC row is apparent performance on the training cohort and is not validation. On the two RFS-

Cohort	Role	n	Brier (IPCW)	Null Brier	Skill
METABRIC_apparent	apparent	1975	0.1718	0.1910	0.100
GSE7390	external	198	0.1988	0.2042	0.026
GSE2990	external	187	0.1731	0.1867	0.072

Table 9: IPCW Brier scores at 60 months. Skill is $1 - B/B_{\text{null}}$; values near zero mean the score barely beats assigning everyone the cohort average.

comparable external cohorts the skill over the null is small: 0.026 in GSE7390 and 0.072 in GSE2990. The score adds a little absolute-risk information beyond the cohort average, not enough to support individual treatment decisions. Decision-curve analysis over threshold probabilities 0.05–0.60 (same IPCW weights; Table 10) shows net benefit above treat-all mainly in the 0.15–0.25 threshold band; at low thresholds the model and treat-all are nearly identical, and at 0.30 and above treat-all goes negative while the model stays positive but small. These

Cohort	NB @ 0.15	NB @ 0.20	NB @ 0.25	NB @ 0.30
METABRIC_apparent	0.134	0.109	0.084	0.069
GSE7390	0.172	0.125	0.059	0.018
GSE2990	0.119	0.091	0.062	0.040

Table 10: Decision-curve net benefit at four thresholds (treat-none is zero by construction). Descriptive audit only; thresholds for care are a clinical, not statistical, decision.

curves are descriptive: they use a transported baseline without cohort refit, assume independent censoring given time, and cannot recommend a clinical threshold. They bound, rather than promote, the score’s decision value.

4.18 Competing-risks descriptive check (cause-specific, not Fine–Gray)

Recurrence is not the only way patients leave the risk set. Coding three states from the committed clinical records (`run_competing_risks.py`, `results/competing_risks.json`; design locked in the same commit as first run): 800 recurrences, 431 deaths without recurrence, 744 censored. Death before recurrence is therefore a material competing event (about one death for every two recurrences). Cause-specific penalized Cox fits on the same clinical+expression features give apparent concordance 0.709 for recurrence and 0.792 for death without recurrence (in-sample, so optimistic; the validated recurrence figure remains the 0.680 cross-validated benchmark). Cause-specific models censor the other event type and answer a different question than the Fine–Gray subdistribution model, which remains not done here: no validated implementation of its time-varying censoring weights exists in this codebase, and a substituted model under the Fine–Gray name would be mislabeled. The descriptive implication stands: absolute recurrence-risk estimates that ignore the 431 competing deaths will overstate five-year recurrence probability, which bounds every calibration number in this paper.

4.19 Gene-family-blocked cross-validation of the 70-gene panel

Which functional families carry the panel’s discrimination? The audit was preregistered before any family arm was fit; the locked design is docs/PREREG_FAMILY_BLOCKED.CV_20260928.md: eight a-priori families covering all 70 genes exactly once, penalized Cox (penalizer 0.05) with the four declared clinical covariates in every arm, five-fold patient cross-validation (seed 42, pooled out-of-fold concordance) plus the fixed seed-42 holdout, and 100 size-matched random gene-set nulls per family (per-draw seeds 20260928+i; run_family_blocked_cv.py, results/family_blocked_cv.json). Baseline pooled-CV concordance 0.6550, holdout 0.6464. Two honest readings follow. First, no single family is load-bearing under the locked rule: the

Family (genes dropped)	<i>n</i>	CV5 <i>C</i>	Δ CV5	Holdout <i>C</i>	Δ holdout	null Δ p95
basal_myoepithelial	10	0.6618	-0.0068	0.6498	-0.0034	+0.0060
cellcycle_apoptosis_signaling	5	0.6598	-0.0048	0.6458	+0.0006	+0.0042
dna_repair_genome_stability	8	0.6576	-0.0026	0.6507	-0.0043	+0.0052
er_luminal	14	0.6536	+0.0014	0.6426	+0.0038	+0.0106
her2_amplicon	2	0.6552	-0.0002	0.6482	-0.0018	+0.0034
hypoxia_adhesion_detox_other	6	0.6542	+0.0008	0.6449	+0.0015	+0.0060
pi3k_akt_mtor	4	0.6541	+0.0009	0.6466	-0.0002	+0.0053
proliferation_mitotic	21	0.6454	+0.0095	0.6486	-0.0022	+0.0075

Table 11: Leave-one-family-out concordance. Δ is baseline minus arm, so positive values mean the family helps. Under the locked rule (CV5 drop > 0.005 AND holdout drop above the size-matched null 95th percentile) no family is load-bearing.

largest cross-validation drop is proliferation_mitotic (+0.0095), but its fixed-holdout drop is -0.0022 (percentile 50 against its size-matched null), a split verdict that echoes the earlier proliferation negative rather than contradicting it. Second, removing basal_myoepithelial raises pooled-CV concordance to 0.6618 and is flagged dispensable-or-harmful under the same locked rule; the effect is small, the holdout does not confirm it, and it is reported as a bound on what the panel needs, not a pruning claim. The panel’s discrimination is distributed across families rather than carried by any one of them.

4.20 Patient-level endpoint harmonization before pooling

The endpoint boundary documented above forbids pooling unlike endpoints; this audit executes the patient-level harmonization itself. It was preregistered before any harmonized endpoint was derived (docs/PREREG_ENDPOINT_HARMONIZATION_20260928.md; raw per-patient clinical pulls and their SHA-256 manifest committed before any outcome was computed). Two strata are locked: HARM-RFS (event = any recurrence, locoregional or distant) and HARM-DMFS (event = distant metastasis or distant relapse only). A cohort enters a stratum only when its raw fields express that stratum’s event; there is no imputation and no proxy mapping across endpoint types. Pooling happens only within a stratum (patient-level bootstrap stratified by cohort, 500 replicates, seed 0); the two strata are never pooled together or compared as if they were one endpoint. The model is the committed transport model - penalized Cox (penalizer 0.05) on the METABRIC 70-gene panel, fit once, with no re-tuning after any harmonized outcome was seen. The preregistration locks no performance gate: this is a methodological audit, and every number below is descriptive. Where a harmonized definition coincides with a committed transport definition the frozen pipeline reproduces the committed value exactly (GSE2990: HARM-RFS 0.6559, HARM-DMFS 0.6648), a determinism check rather than a new result. The pooled HARM-DMFS concordance 0.6477 (0.6191, 0.6755) sits above the

Stratum	Cohort	Patients	Events	Mapped genes	C (95% CI)
HARM-RFS	GSE2990	187	67	63	0.6559 (0.5778, 0.7313)
HARM-DMFS	GSE2990	179	40	63	0.6648 (0.5685, 0.7467)
HARM-RFS	GSE7390	198	91	63	0.5997 (0.5360, 0.6513)
HARM-DMFS	GSE7390	198	62	63	0.6352 (0.5711, 0.6954)
HARM-DMFS	GSE11121	200	46	63	0.6979 (0.6088, 0.7771)
HARM-DMFS	GSE20685	327	83	69	0.6270 (0.5638, 0.6907)
HARM-RFS	GSE20685	307	83	69	0.6033 (0.5397, 0.6804)
HARM-DMFS	GSE25066	508	111	63	0.6400 (0.5912, 0.6821)
HARM-RFS	<i>pooled</i> (3 cohorts)				0.6165 (0.5725, 0.6577)
HARM-DMFS	<i>pooled</i> (5 cohorts)				0.6477 (0.6191, 0.6755)

Table 12: Harmonized transport concordance by stratum. Within a stratum every cohort carries the same patient-level event definition; pooled rows are cohort-stratified bootstrap intervals, not cross-endpoint averages. GSE11121 and GSE25066 have no any-recurrence field in their committed raw pulls and enter HARM-DMFS only; GSE25066 contributes distant-relapse (DRFS) fields, which the locked definition treats as the distant-event endpoint.

pooled HARM-RFS concordance 0.6165 (0.5725, 0.6577), but the two strata draw on different cohort sets and their intervals overlap; the preregistration forbids reading this as an endpoint comparison. Mapped-gene counts differ by platform (63 of 70 on GPL96 cohorts, 69 of 70 on GPL570), so cross-cohort contrasts also carry a platform-coverage difference. Per-cohort intervals overlap heavily; no cohort ordering is claimed.

5 Discussion

A penalized linear Cox model on a 70-gene panel is the ceiling in this data regime: across eight controlled comparisons (deep Cox, GBT survival, multimodal methylation/mutation/CNA integration, engineered features), no alternative exceeded it, and several were significantly worse. The transport experiments sharpen that verdict: the same frozen coefficients have higher point estimates than indirect signature comparators in some cohorts and lower estimates than a continuous GGI and nodal stage in others. These are not direct same-patient comparisons with marketed MammaPrint or Oncotype DX scores, nor do endpoint-mismatched cohorts establish a pooled win. On the untouched native-DRFS cohort GSE17705 the frozen panel beats neither chance nor a nodal-only comparator (0.509 [0.437, 0.578] versus 0.618) - the sharpest external boundary in this paper. External replication, not internal cross-validation, is the honest arbiter of a prognostic claim; failed discovery arms remain documented, with their consequences summarized here rather than presented as positive findings. Limitations: the panel is expression-only, cohorts differ in treatment era and endpoint definition, and the GSE20685 result suggests clinical covariates still carry signal the panel misses.

6 Reproducibility

All code (51 hermetic tests), cached API snapshots, result JSON files, and figures are in the project repository; every number in this paper is regenerable from `results/*.json`. The audit-corrected resource roster has at most 36 distinct science/data tools before direct-use scrutiny and does not meet the 40-tool gate. Its 3,797 patient/sample identifiers are nested records, not independent study datasets. The 120-dataset gate is conditional on direct record-level use and uniqueness; the source-study count is nine and does not meet it.

Cohort	Stratum	Derivation from committed raw pull
GSE2990	HARM-RFS	event.rfs/time.rfs verbatim (years; unit-free C-index)
GSE2990	HARM-DMFS	event.dmfs/time.dmfs verbatim (years)
GSE7390	HARM-RFS	e.rfs/t.rfs verbatim (GEO 'key: value' characteristics cells, key-exact, prefix stripped at parse)
GSE7390	HARM-DMFS	e.dmfs/t.dmfs verbatim (same key-verbatim extraction)
GSE11121	HARM-DMFS	t.dmfs/e.dmfs verbatim (months; key-verbatim extraction as GSE7390)
GSE11121	HARM-RFS	EXCLUDED: no any-recurrence field in committed raw pull
GSE20685	HARM-DMFS	event.metastasis + follow_up_duration (years) verbatim (key-verbatim extraction)
GSE20685	HARM-RFS	any recurrence = regional_relapse OR event.metastasis, key-verbatim (20 of 327 pulled patients carry regional_relapse: NA, no usable value - excluded from this stratum only, leaving n=307); same follow-up field
GSE25066	HARM-DMFS	drfs_1_event_0_censored + drfs_even_time_years verbatim (distant relapse; key-verbatim, 198 of 508 rows recovered from drifted columns)
GSE25066	HARM-RFS	EXCLUDED: no any-recurrence field in committed raw pull (neoadjuvant pCR cohort)

Table 13: Derivation audit: every stratum assignment traces to verbatim fields in the committed raw pulls. Exclusions are recorded with reasons; 20 of 327 GSE20685 patients carry regional_relapse: NA and leave the HARM-RFS stratum only. TCGA stays out of both strata: its committed analysis is a case-control early-recurrence comparison, not a patient-level survival transport.

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